

6.04 Centre of Mass			
6.04a	Centre of mass		a) Understand and be able to apply the principle that the effect of gravity is equivalent to a single force acting at the body's centre of mass. <i>Includes understanding that, in terms of linear motion, a rigid body may be modelled by a particle of the same mass at its centre of mass.</i>
6.04b			b) Be able to find the position of the centre of mass of a uniform rigid body using symmetry, for example a rectangular lamina.
6.04c			c) Be able to determine the centre of mass of a system of particles or the centre of mass of a composite rigid body. <i>Questions may involve any of the rigid bodies listed in the Formulae Booklet, but will be limited to compound shapes such as a uniform L-shaped lamina or a hemisphere abutting a cylinder with a common axis.</i> <i>Includes composition by addition or subtraction, for example a rectangular lamina with a semicircle attached to one side, or a rectangular lamina with a semicircle removed.</i>
6.04d			d) Be able to use integration to determine the position of the centre of mass of a uniform lamina or a uniform solid of revolution.
6.04e	Rigid bodies		e) Be able to solve problems involving the equilibrium of a single rigid body under the action of coplanar forces. <i>e.g. Suspension of a rigid body from a given point or problems involving the toppling or sliding of a rigid body placed on an inclined plane.</i> <i>May include rigid bodies which are hinged to a surface.</i> <i>[Hinged bodies are excluded.]</i>

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
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